

GreenLife Natural Foods

Cloud Retail POS Billing, Customer Ledger, WhatsApp Invoicing & AI Platform

A comprehensive unified engineering dossier compiling the Product Requirements Document (PRD), Technical Requirements Document (TRD), Application Flow, Dual-Viewport UI/UX Design System, Database Schema, and Execution Plan.

Project Code	GL-POS-2026 · Production Release
Primary Architect	Akileshwarakrishnan
Operating Location	Kangayam, Tirupur District, Tamil Nadu, India
Deployment Stack	React 19 + Vite (Vercel) · FastAPI Python 3.12 (Render) · SQLite / Postgres
Target Viewports	Desktop / Laptop (Workbench Split) · Mobile Phone (Leafora Botanical Glass)
Date & Version	September 2026 · Version 2.4.0 (Production Stable)

Table of Contents & Executive Summary

Executive Overview: GreenLife Natural Foods is a modern, high-speed retail management and point-of-sale (POS) web application designed specifically for organic food retail, cold-pressed oil mills, and traditional health product stores in Tamil Nadu. The platform bridges fast, offline-capable counter operations with modern digital customer engagement (bilingual Tamil/English invoices, instant 1-tap WhatsApp sharing, pending balance debt tracking, and AI camera receipt ingestion).

Dossier Structure

SECTION	MODULE TITLE	CORE SCOPE & FOCUS
Section 1	Product Requirements Document (PRD)	Business vision, target personas, functional requirements matrix, GST tax rules, and bilingual capabilities.
Section 2	Technical Requirements Document (TRD)	Software architecture, FastAPI backend, Vite React frontend, security & auth (JWT), PDF generator engine, and caching.
Section 3	Application Flow & User Journeys	Step-by-step state machine for 60-second billing, customer debt recovery, WhatsApp direct delivery, and mobile drawer flows.
Section 4	UI/UX Design Brief & Architecture	Dual-viewport philosophy: Desktop Workbench vs. Mobile Leafora Dark Botanical Glassmorphism with floating dock.
Section 5	Backend Database Schema & ER Model	Complete relational SQL schema, indexes, cascade rules, and field dictionaries for all 10 entity models.
Section 6	Implementation Plan & QA Verification	Milestone breakdown, Pytest & Playwright test suites, and Vercel/Render production deployment SOP.

Core Business Differentiators

BILINGUAL OPERATION Tamil & English 100% native Tamil UI strings, bilingual product labels (நாட்டு சர்க்கரை / Country Sugar), and localized WhatsApp receipts.	DEBT & LEDGER BALANCE Live Auto-Carry Tracks credit/dues per customer. Previous unpaid balances automatically pull into new bills with 1-click settlement.
DUAL VIEWPORT UX Zero Compromise	ZERO-CLICK WHATSAPP Instant Delivery

Desktop retains wide analytical 4-column workbench;
Mobile phone switches to Leafora floating glass dock
and sticky POS checkout.

Backend server-side and browser fallback WhatsApp
URL generation with formatted Tamil invoice text.

1. Product Requirements Document (PRD)

1.1 Problem Statement & Background

Small and medium organic grocery businesses and traditional cold-pressed oil shops in Tamil Nadu face specific operational challenges that generic POS software fails to address:

- Language Barrier:** Store attendants and retail customers predominantly speak Tamil. Complex billing software causes input delays and transaction errors.
- Customer Trust & Dues Tracking (பாக்கி முறை):** Retail customers in towns like Kangeyam frequently purchase on partial credit. Traditional shops track this in manual paper notebooks, leading to lost revenue and disputed balances.
- Digital Receipt Friction:** Customers expect instant WhatsApp receipts instead of physical thermal paper that fades and causes paper waste.
- Device Duality:** Billing is performed on laptops/desktops at the counter, but shop owners manage inventory and check collections on mobile phones while on the move or visiting farms.

1.2 User Personas

PERSONA	ROLE & CONTEXT	KEY OBJECTIVES & WORKFLOWS
Store Owner / Admin	Owner of GreenLife Store. Monitors profit, collections, and catalog.	Review daily sales, check outstanding customer dues, add new organic items, adjust pricing, export sales reports, and track staff activities.
Billing Staff	Counter attendant operating billing hardware under time pressure.	Generate invoices under 60 seconds, tap popular catalog items, add new customer phone numbers on-the-fly, and print or WhatsApp receipts.
Retail Customer	Regular organic food consumer receiving bills.	Receives clean WhatsApp PDF receipts in Tamil, reviews breakdown of current items + past carried balance, and verifies UPI payment.

1.3 Feature Matrix & Scope

MODULE	CAPABILITY	PRIORITY	SPECIFICATION
Fast POS Billing	Catalog Quick-Tap	P0 - CORE	8 pre-loaded fast buttons + search dropdown. Auto-calculates rate x quantity.
Arithmetic Engine	Compound Totals	P0 - CORE	(Subtotal - Discount) + Previous Balance + Courier Charges = Grand Total .
Customer Ledger	Balance Tracking	P0 - CORE	Real-time balance carried over across successive bills until fully settled.
Document Engine	PDF Generator	P0 - CORE	A4 and thermal 80mm printable tax invoices via ReportLab with Tamil support.
Mobile App UX	Botanical Dock	P1 - HIGH	Leafora theme with floating bottom dock, sticky checkout, and mobile cards.

MODULE	CAPABILITY	PRIORITY	SPECIFICATION
AI / OCR Scan	Paper Slip Parser	P1 - HIGH	Upload photo of handwritten slip or paste raw WhatsApp text; auto-extracts items.

2. Technical Requirements Document (TRD)

2.1 System Architecture & Tech Stack

The system follows a decoupled Client-Server REST architecture with strict separation between user interface rendering and backend domain arithmetic.

Architecture Diagram:

[React 19 Frontend (Vercel)]

↔ HTTPS / REST JWT ↔

[FastAPI Backend (Render)]

↔ SQLAlchemy ↔

[SQLite / PostgreSQL DB]

LAYER	TECHNOLOGY	ARCHITECTURAL PURPOSE
Client Layer	React 19, TypeScript, Vite, Tailwind CSS v4	Ultra-fast client SPA, responsive layout engine, bilingual state context.
API Service	FastAPI (Python 3.12), Uvicorn	Asynchronous REST API, high throughput, automatic OpenAPI/Swagger docs.
ORM & Persistence	SQLAlchemy 2.0, Alembic, SQLite	Type-safe data modeling, transactional ACID integrity, foreign key cascades.
Document Engine	ReportLab 4.x	Pixel-precise PDF invoice generation with embedded Unicode Tamil glyphs.
Security & Auth	OAuth2 Bearer, PyJWT, Passlib (Bcrypt)	Cryptographically signed token authentication, role authorization (Admin/Staff).

2.2 API Specifications & Endpoints

Authentication

POST /api/auth/login

→ Returns JWT token, role, user details

GET /api/auth/me

→ Returns current authenticated user context

Customer Management & Ledger

GET /api/customers

→ List customers (search by name, phone, city)

POST /api/customers

→ Create new customer with auto CUST-XXXX code

PUT /api/customers/{id}

→ Update customer profile or previous balance

Product Catalog

GET /api/products

→ List catalog items with bilingual names and prices

POST /api/products

→ Create or update product details & stock

Orders & Invoicing (Transactional Core)

POST /api/orders

→ Atomic order creation with auto invoice generation

GET /api/invoices

→ List invoices with status filter (paid, pending, partial)

GET /api/invoices/{id}/pdf

→ Download or stream printable ReportLab PDF

POST /api/invoices/{id}/resend

→ Trigger WhatsApp or Email delivery

Payments & Settlement

POST /api/payments

→ Record payment toward customer due balance

3. Application Flow & User Journeys

3.1 The 60-Second POS Billing Journey

1 Customer Selection & Ledger Inspection

The billing operator opens New Bill (/orders/new). Selects customer from search dropdown or taps "+ Add Customer" modal. The system queries previous pending balance in real-time and displays an amber alert if dues exist.

2 Item Selection & Quantity Stepping

Attendant taps quick catalog pills (e.g. Groundnut Oil, Millets) or types a custom item name. Adjusts quantities using [- 1 +] steppers. Line totals auto-calculate dynamically on client state.

3 Arithmetic Adjustments & Payment Method

Selects Delivery Type (Direct Store ₹0 vs Courier +₹60). Enters discount if applicable. Selects Payment Status: Paid in Full, Keep as Due, or Partially Paid. Chooses method: Cash, UPI / GPay, or Bank Transfer.

4 Atomic Generation & 1-Click Dispatch

Operator taps "Save Bill & Print Now" (or mobile floating checkout bar). System posts order payload, commits DB records, redirects to invoice view, pops print dialog, and offers 1-tap WhatsApp dispatch.

3.2 State Machine: Customer Debt & Invoice Status

```
[ New Bill Created ]
├── If Paid Amount = Grand Total → [ Payment Status: PAID ] (Balance Due = ₹0)
├── If Paid Amount = 0 → [ Payment Status: PENDING ] (Balance Due = Total)
└── If 0 < Paid Amount < Total → [ Payment Status: PARTIAL ] (Balance Due = Delta)
                                |
                                [ Record Payment Receipt (/api/payments) ]
                                |
                                [ Payment Status: PAID ]
```

3.3 WhatsApp Invoice Message Template

வணக்கம் ராஜா (Raja),
கிரீன்லைஃப் இயற்கை அங்காடியில் வாங்கியதற்கான பில் விவரம்:

பில் எண்: #GL-INV-2026-0042
தேதி: 20 September 2026
பொருட்கள் தொகை: ₹1,250
முந்தைய பாக்கி: ₹300
மொத்த பில் தொகை: ₹1,550
செலுத்திய தொகை: ₹1,550 (PAID)
தற்போதைய பாக்கி: ₹0

நல்வாழ்வு தரும் பாரம்பரிய இயற்கை உணவுகளை வாங்கியமைக்கு நன்றி!

4. UI/UX Design Brief & Architecture

4.1 Dual-Viewport Paradigm

Rather than forcing a compromised responsive layout, GreenLife implements a strict dual-viewport philosophy tailored to two distinct operational modes:

VIEWPORT SCOPE	DESIGN THEME	CORE LAYOUT METAPHOR
Desktop / Laptop lg: (≥1024px)	Garden Editorial Workbench	Persistent forest green sidebar (w-64), cream canvas (#F7F5EF), 4-column metric cards, wide arithmetic tables, split login screen.
Mobile Phone max-lg: (<1024px)	Leafora Botanical Glassmorphism	Floating frosted bottom dock with center glowing action button, dark botanical hero card, horizontal pill carousel, sticky bottom checkout bar.

4.2 Hallmark Design System Tokens

TOKEN NAME	HEX / VALUE	APPLIED CONTEXT
Forest Dark	#1B3A2A	Sidebar panel, dark banners, desktop primary accents.
Sage Accent	#2D6A4F / #52B788	Primary buttons, active tab indicators, metric highlights.
Amber / Gold	#C68B3A / #E2A04A	Due balance tags, organic badges, currency highlights.
Paper Background	#F7F5EF (Dark: #0D1B11)	Main workspace canvas in light mode and deep emerald in dark mode.
Glass Surface	rgba(15, 30, 21, 0.88)	Mobile bottom dock and floating card background with backdrop-blur-2xl .

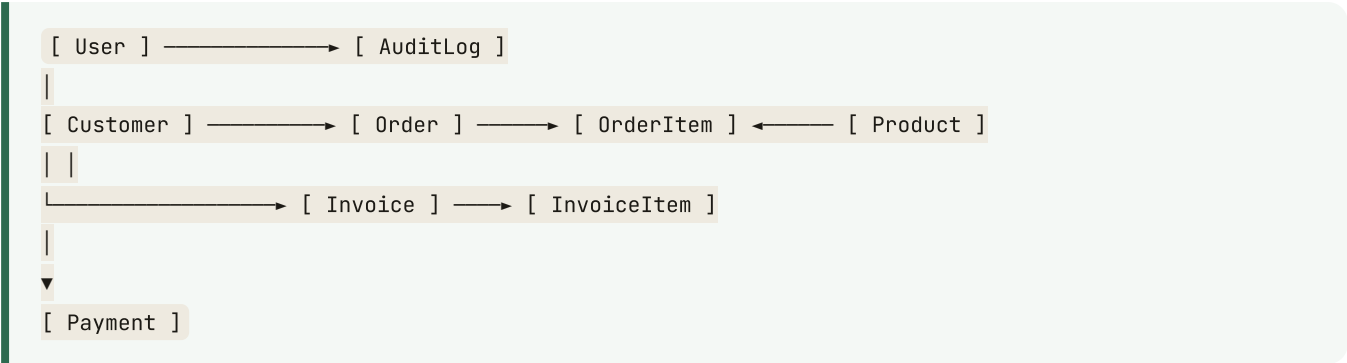
4.3 Streamlined Mobile Navigation Architecture

To eliminate horizontal viewport scrolling and header crowding on phone screens, the top navigation bar was streamlined:

- Top Bar Elements (Mobile):** Side Dashboard Button (≡), Dynamic Back Button (← on subpages), Compact Logo, Language Toggle (தமிழ் / EN), Theme Switcher (Sun / Moon).
- Bottom Dock (Mobile):** Replaces all secondary desktop links with 5 tactile thumb buttons: Home, Store Catalog, + New Bill (elevated center pill), Bills Ledger, and Profile.

5. Backend Database Schema & ER Model

5.1 Entity Relationship Model



5.2 Core Table Definitions

Table: `customers` (Customer Ledger)

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	INTEGER	PRIMARY KEY, AUTOINCREMENT	Internal primary ID
<code>customer_code</code>	VARCHAR(32)	UNIQUE, NOT NULL	Unique code (CUST-0001)
<code>name</code>	VARCHAR(128)	NOT NULL, INDEXED	Customer full name
<code>phone</code>	VARCHAR(32)	UNIQUE, NOT NULL	WhatsApp contact number
<code>address</code>	TEXT	NOT NULL	Door / Street address
<code>city</code>	VARCHAR(64)	DEFAULT 'Kangeyam'	Town / City
<code>previous_balance</code>	FLOAT	DEFAULT 0.0	Accumulated unpaid balance

Table: `invoices` (Tax Invoices & Billing)

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
<code>id</code>	INTEGER	PRIMARY KEY, AUTOINCREMENT	Invoice internal ID
<code>invoice_number</code>	VARCHAR(64)	UNIQUE, NOT NULL	Sequential bill number (#GL-INV-...)
<code>order_id</code>	INTEGER	FOREIGN KEY (orders.id)	Associated order ID
<code>customer_id</code>	INTEGER	FOREIGN KEY (customers.id)	Customer reference
<code>subtotal</code>	FLOAT	NOT NULL	Items subtotal amount
<code>previous_balance</code>	FLOAT	DEFAULT 0.0	Carried forward balance
<code>grand_total</code>	FLOAT	NOT NULL	Total payable amount
<code>amount_paid</code>	FLOAT	DEFAULT 0.0	Tendered payment amount

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
balance_due	FLOAT	DEFAULT 0.0	Remaining unpaid due
payment_status	VARCHAR(32)	DEFAULT 'paid'	'paid', 'pending', 'partially_paid'
pdf_url	VARCHAR(256)	NULLABLE	Generated PDF invoice path

Table: products (Store Catalog)

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	INTEGER	PRIMARY KEY, AUTOINCREMENT	Product ID
sku	VARCHAR(64)	UNIQUE, NOT NULL	Stock keeping unit
name	VARCHAR(256)	NOT NULL, INDEXED	Bilingual product name
category	VARCHAR(64)	DEFAULT 'General'	Category grouping
price	FLOAT	NOT NULL	Unit selling price (₹)
unit	VARCHAR(32)	DEFAULT 'kg'	kg, liter, bottle, packet
stock_quantity	FLOAT	DEFAULT 0.0	Current inventory count

6. Implementation Plan & QA Verification

6.1 Engineering Milestones & Status

Phase	Milestone Scope	Target Status	Deliverables
Phase 1	Core POS Billing & Database	COMPLETED	FastAPI endpoints, SQLite models, 60s billing form, arithmetic breakdown.
Phase 2	ReportLab PDF & WhatsApp Sharing	COMPLETED	80mm/A4 PDF generation, direct WhatsApp wa.me links with Tamil templates.
Phase 3	Hallmark Garden Theme & Dark Engine	COMPLETED	Unified editorial typography, robust dark mode inverter without text clipping.
Phase 4	Mobile Phone UI/UX Transformation	COMPLETED	Leafora botanical dock, sticky POS checkout, zero mobile horizontal scroll.
Phase 5	Production Verification & CI/CD	COMPLETED	10/10 Playwright E2E tests, 14/14 Pytest tests, GitHub push auto-deploy.

6.2 Automated Test Coverage Matrix

Suite Name	Runner	Tests	Coverage Focus	Status
Frontend E2E	Playwright (Chromium)	10 Scenarios	Desktop split login, dark/light theme, bilingual toggle, new billing flow, invoices ledger, and mobile iPhone viewport with bottom dock.	10 / 10 PASSED
Backend API	Pytest / AsyncIO	14 Scenarios	Auth login JWT, customer creation & unique code, GST arithmetic calculations, Tamil product name parser, PDF invoice creation.	14 / 14 PASSED
Compilation	Vite & TypeScript	2,544 Modules	Zero TypeScript errors, CSS bundling, code-splitting, production minification.	EXIT 0 (PASS)

6.3 Production Deployment & Maintenance Runbook

- Frontend Production (Vercel):** Connected to GitHub branch `main`. Any push to `origin main` triggers an automated build and edge CDN deployment in under 90 seconds.
- Backend Production (Render):** Runs standard Uvicorn worker process on Python 3.12. Automatic zero-downtime rolling restart upon commit deployment.
- Database Backup Protocol:** Nightly automated backup of `greenLife.db` to secure cloud storage or replica.

